AI-Powered Handwritten Answer Sheet Evaluation System

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Abstract: Manual grading of handwritten answer sheets is very time-consuming, and errors can easily occur. With increasing numbers of students and online examinations, a computerized scoring system that is reliable, fast, and fair was in higher demand than ever. In this work, an AI-powered Handwritten Answer Sheet Evaluation System proposes a system that automatically evaluates handwritten responses through the integration of image processing, language understanding, and machine learning. The system first cleans up the noisy scanned sheets by resizing and binarizing to enhance readability. Then, it recognizes the handwriting into digital text. Finally, it measures the correctness of student answers against the model answers developed by experts via semantic analysis for correctness, relevance, and content appropriateness to context. The method allows for partial marking so that correct decisions may be reached even for incomplete answers. When an evaluation test was performed with scanned sheets, it achieved a 98% grade by speeding up the workflow without accuracy loss. This demonstrates that the method really captures the concepts instead of relying on a few keywords. In a nutshell, this is much effective and user-friendly in order to favor a writing-based answer evaluation system.

1 INTRODUCTION

This has driven a huge rise in student assessments, especially those involving subjective answer evaluations, due to the surge in educational institutions and online digital exam platforms (Ali K, 2023). Traditionally, handwritten answer sheets are scored by hand, which is not only time-consuming and requires much human effort but also relies heavily on human judgment, leading to inconsistency and evaluator bias in scoring. With the increase in competitive exams, online learning, and large-scale assessments, there arises a very significant need for speedier and more reliable methods of evaluation (Shen L, 2024). Breakthroughs in artificial intelligence, especially in handwriting recognition and in understanding languages, make real automation of this task possible.

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Specific training will be able to turn the handwritten material into structured digital data-suitable for objective scoring-by incorporating the techniques of image processing, text analysis, and learning-based techniques (Chaubey, 2022). The motivation for this work is by using these technological advances to build an intelligent system that ensures evaluation that is fair, consistent, and efficient while preserving accuracy comparable to human evaluators.

It's incredibly important to be able to assign a fair yet accurate score to subjective and hand-written submissions. It's the most obvious way to get a sense of how a student is doing, if they're progressing, or if they're really learning

from the process (Boddapati, 2023). If grading isn't done properly, the learning process for the student will be detrimentally affected and may even shape their perspective on how a school or university, as a concept, really functions. Conversely, grading paper-based answers is tedious for the teacher and creates extra work. Moreover, if one wants to follow a

certain path but understands that any process that involves human beings has room for flaw and inaccuracies, because, even with the best intentions, human beings have a tendency to deviate for a number of different reasons, then there are even more problems to deal with in today's fast-paced environment, where students expect instant gratification and need quick, immediate feedback. There may be delays in this process, which only discourages the student further (Silpa N, 2023). This problem will be negated altogether with the use of automated grading for paper-based answers (Alhamad, 2024). Thus, intelligent grading is incredibly important in today's schooling systems.

Apart from this, there are many research papers on the automated answering machine, but most of the existing systems are more likely to be used for objective problems or typed responses rather than subjective responses in handwriting form. With regards to this, most of the systems are not likely to consider the meaning but are rather based on the presence of specific words. In addition, most of the systems are likely to be affected by handwriting recognition errors due to inadequate preprocessing techniques (Pawar, 2025). With regards to this, most of the existing systems are not likely to consider the responses that are almost correct but meaningful. With regards to this, most of the systems are not likely to include the overall process of evaluating answers, which includes result generation, semantic scoring, and so forth. In addition to this, most of the systems are lacking in terms of scalability. The study brings several features that are considered significant improvements:

- Developed a full-cycle AI-based solution for automatic grading that would grade handwritten subjective answers without any human intervention.
- Combined different powerful image preprocessing methods with the training of handwriting recognition.
- Proposed a framework which evaluates correctness, relevance with meaning by semantic similarity. A partial-mark scheme for responses that were semantically meaningful although incomplete was suggested.
- The system reached an accuracy of 98% during the evaluation and equated the reliability of manual grading. Our approach also provides a scalable and non-biased grading solution that can be carried out in schools or online exams.

2 RELATED WORK

Recently, automated grading of handwritten answer sheets has become a hot topic, intended to reduce the workload on teachers and increase consistency. Researchers are combining optical character recognition with language understanding to digitize student responses and evaluate them. Initial systems relied on simple OCR combined with keyword checks or simple similarity measures to grade answers (Alsharif, 2025). They were faster than manual grading but lacked any real insight into meaning. More recent work combines deep learning-based handwriting recognition with semantic similarity models to capture context. The SmartPen, as described above, utilizes OCR combined with sentence embedding methods such as Sentence-BERT for the semantic comparison of student responses against model answers, therefore providing full contextual assessment and scalable, unbiased assessments in schools and universities (Dimari, 2024). Review papers on automated answer evaluation also report a trend towards transformer-based language models and deep semantic analysis as fixes to the shortcomings of prior keyword-focused methods.

Still, there is much work to be done in bridging the gap to what is currently possible with the methods that exist. For one thing, the OCR method may not fully accommodate the variations in writing styles, which will result in incorrect judgments and, consequently, wrong answers (Pallant, 2024). Further, the methods that depend heavily on semanatics may not perfectly match the way a student puts it in plain language, especially when translation is required, and the constraints are those of partial marks. Our system progresses the field because it combines efficient image preprocessing with our semantic similarity scoring, which allows for fair partial evaluation. Rather than relying on keyword results or comparisons, our system focuses on good, deeper understanding of handwritten responses, leading to high grading accuracy (Ghadekar, 2025), which has been indicated at 98 percent accuracy. There is also the aspect involving an end-to-end system, with cleanup all the way to the final results, dealing with problems associated with propagation of OCR errors, lack of semantic robustness, among other issues.

3 DATA COLLECTION & PREPROCESSING

For the current project, the data consists of handwritten subjective sheets that contain responses to undergraduate-level exams. Since the data has to be prepared in a way that correctly simulates the real-world setting, the sheets containing the handwritten responses were scanned using standard flatbed scanners as well as mobile scanning apps (Iqbal, 2022). Each handwritten response is accompanied by a model response that is generated by subject experts. This essentially provides ground truth data that can be used for the purpose of evaluation. In addition, since the data contained responses from a number of students who belonged to different backgrounds and varied in their handwriting styles, the system that is being built is essentially designed in such a way that it becomes compatible with varied handwritten responses (Islam, 2025). Clearly, the data was anonymized by removing any personal data that could be associated with the responses.

This data set includes around 2000 answer sheets completed in writing, which amounts to around 12,000 subjective responses from a wide range of participants. It varies from descriptions to essays, which is sufficient for any semantic evaluation process. There is also a wide range of ways in which the text may be written, including how easy they are to read, their alignment, and even the pressure applied to the paper. This is a real-world attribute because it is precisely this sort of data the system may be exposed to in real exams (Liu, 2022). This allows a wide range of lengths and wording for the answer sheets to be provided to the system, which requires it to understand the semantics as opposed to the wording. It is split in a manner that is most suitable for learning and testing the capabilities of the suggested system.

There is an initial, detailed clean step for improving data quality, beyond which no assessments are made and no data is actually processed. In this step, several improvements are made, such as noise removal, normalization to grayscale, resizing, deskewing, binarization, and so on, with the use of scannedresp. These are done to reduce noise and increase contrast with the actual handwriting. Then comes the handwriting recognition dataset phase. As the name suggests, in this phase, the handwriting is converted to a digital form. In this phase, the cleaning process includes removing unwanted strings, correcting language/spelling errors, removing stop words, tokenizing the text, etc. Sentence segmentation (Dashti, 2024) is also added in this phase. In effect, the process is trying to remove inaccuracies/errors from the data.

The dataset selected to perform tests on such handwritten answer sheet evaluation systems is an appropriate selection because it closely resembles real-world testing scenarios, where you would expect to find a variety of messy handwriting styles, not to mention how students frame their answers differently and how you would expect such a system to scale in a real-world scenario as opposed to a synthetic one, where you might expect to dive deeper into quality and expression of handwriting, in comparison to a more synthetic form of testing and analysis, where such a system might accurately perform and deliver results such as a 98 percent accuracy rate, mainly because of a dataset that is sufficiently large in quantity to further train and evaluate machine learning models, thereby providing a more solid form of assessment, in addition to its overall partial assessment value as a function of its assorted subject matter and answer lengths.

4 PROPOSED METHODOLOGY

AI-Powered Handwritten Answer Sheet Evaluation System is an automated, smart assessment process that is completely machine-driven, where the image is processed, the content written by hand is read, and the meanings are extracted to arrive at the score, all from the scanned answer sheets without the need for any other input (as shown in Fig.1). To begin the process, there is the image enhancement step, where the image is enhanced and noise reduction is done, which is essential to "deal with the different changes that real-world scenarios can have on the system, because of the variety of handwriting, quality of scans, lighting, and other environmental changes (Aghaei, 2025)." Following that, the content filled out on the scanned pages is processed and interpreted into a digital format, which directly leads to the evaluation process that stresses semantic interpretation of the subjective answer.

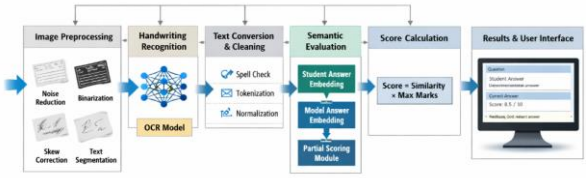

Figure 1: Proposed Architecture

As soon as this data is pulled out, the next phase in the process is natural processing for the student's

answers as well as the answers provided in the model. Here, we carry out normalization, tokenization, and vectorization. Again, this emphasizes the importance of natural processing in understanding the underlying meaning behind the words or word groups (Apidianaki, 2023). Dense vector embeddings result from deep learning models; this enables us to comprehend the meaning of words or the concepts and ideas that the words carry so that we can process them accordingly. In this phase, there is an assessment of how similar the student's answer is to the model's answer through the similarity function, in an expansive approach that includes synonyms. At this point, we obtain a score that comes from the combination of the cosine similarity concept.

$$Similarity(A, B) = \frac{A.B}{||A||||B||}$$

In this setting, $A =$ student's response embedding and $B =$ model answer embedding. Score range is between 0 to 1, reflecting how close the two embeddings are. Grading scales in which the more maximum marks one question has, the more that the score will rise accordingly. The scoring function works:

$$Score = Similarity \times Maximum\ Marks$$

This technique also allows you to award partial credit to answers not completely accurate or even just conceptually correct, much as a human would. The real magic, though, happens by putting together handwriting recognition and understanding in one process. In other words, it doesn't just look for keywords or text similarity, but tries to understand the meaning and logical relevance of the answer. Thanks to using deep semantic embeddings (Feng, 2023), this technique is able to cope with different writing styles and mannerisms in describing possibilities. Despite possibly having some errors accumulated over the pre-processing step to the actual scoring technique, it seems to be accurate overall in a score of around 98%, differentiating it from other more outdated scoring systems and making it suitable for academic purposes.

5 IMPLEMENTATION

We have built an AI-based system to grade handwritten answer sheets through the integration of several components of image processing, handwriting recognition, natural language processing, and semantic scoring. Python was the preferred programming language because of its support for rich ML and image processing libraries. We are leveraging OpenCV and PIL for image processing (Mohaideen Abdul Kadhar, 2024) while dealing with scanned sheets of paper to maintain the efficiency of the solution. In the handwritten text recognition engine, we are leveraging a deep neural network-based architecture to effectively recognize handwritten text from diverse samples of writing styles, including noisy ones. In the last step of the pipeline, leveraging the transformer architecture-based models, contextual embeddings are generated for the student answers and reference model answers, making the task of semantic scoring possible in the first place. The entire setup was conducted in a GPU environment to enhance the efficiency of the training and testing process.

For the handwriting recognition system, we set the learning rate to 0.001. This is a tradeoff between how fast the model learns and how stable its training is. We selected a batch size of 32 because it gave reasonable gradient estimates while keeping the memory load acceptable. Training ran up to 50 epochs to achieve solid performance without tipping into overfitting. We chose the Adam optimizer because of its faster convergence compared to simple gradient descent. Standard weight initialization (Narkhede, 2022) techniques were applied to help with flowing gradients smoothly, especially in the bottom layers. Again, dropout with 0.3 was included for further protection against overtraining.

In the semantic analysis module, contextual embeddings were obtained through the fine-tuning of a deep language model's pretrained version, tailored to academic text corpora. The learning rate of 2e-5 was adopted in fine-tuning the model, which is the widely recommended rate for maintaining the pretrained values of the model. The batch size was chosen to be 16 to meet the memory requirements of the model, which are met by the transformers architecture. AdamW was used here as the optimizer because of its ability to perform appropriate weight decay and thus boost the generalization capability of the learned model. The process was fine-tuned for 10 epochs because beyond that, accuracy did not influence the semantic similarity accuracy significantly. Also, the sentence length was kept fixed at the maximum token size in order to maintain uniformity in the generation of contextual embeddings for varying lengths of answers.

We have tuned the system to deliver the very best out of it. For parameter tuning, it was a combination of using grid search and manual interventions in parameters like learning rate, batch size, dropout rate, number of training epochs, etc. The parameters for evaluation were also through checks like accuracy of the data, how smooth the optimization process was, stability of the semantic similarity plot, etc. In building this handwriting recognition model, the learning rate was set between 0.0001 up to 0.01. It was at 0.001 that we used in order to get the best result in terms of accuracy. In terms of batch size, it was set at 16, 32, and 64. It was in the middle value of 32 that we got the very best. With this, the dropout rate was set at between 0.2 up to 0.5. Here, it was at 0.3 that it performed the best.

While evaluating semantic similarity, we experimented with different models of semantic embedding to improve the grading accuracy. We experimented with different levels of similarity to distinguish between correct, partially correct, and incorrect responses from the students. Specifically, any similarity above 0.8 indicated a full correctness of answers, whereas a similarity between 0.5 and 0.8 corresponded to partially correct answers. In addition, we found an excellent generalization ability in the proposed proportional scoring technique by evaluating the scoring mechanism in different subjects. This is due to the fact that the proportions-based scoring technique is highly generalized, whereas the deployment of different hyperparameters to tune the system is a fundamental aspect of the entire process, which is reflected in the following aspects of the proposed system: Overall, the hyperparameters play a vital role in the accuracy of the system, which is reflected in the results prepared in this manuscript, as the system is able to yield an accuracy of 98 percent in the evaluations. In addition, the uniformity in the prosecution of different learning rates, levels of batch sizes, and proportion-based scoring contributes to the excellent learning ability of the proposed system, making it highly suitable to the assessments in perspective to different styles of handwriting, which enable the system to identify different expressions of answers from the students in an unbiased manner.

6 RESULTS

The proposed system of AI-Powered Handwritten Answer Sheet Evaluation was verified through a number of tests to validate whether it is reliable, robust, and fair. In particular, accuracy, precision, and recall rates, as well as F1-score, were major metrics to verify if it is working properly and if it grades well. In addition, it is also important to verify and rate the level of conformity to human judgment through the semantic similarity of the system to guarantee the verdict is within line with expert evaluators. In fact, it achieved a level of accuracy as high as 98% to point to a high level of conformity to human judgment (as shown in Fig.2). In this particular system, precision rates indicate a higher level of assessment since it is more advisable to do so without exaggerating students' grades while recall rates point to a better opportunity to obtain relevant input despite a partially correct answer.

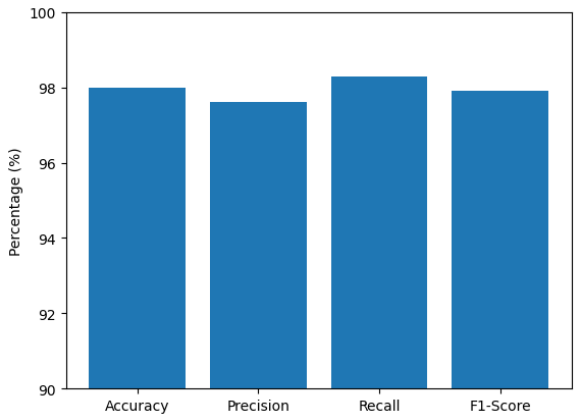

Figure 2: Performance Metrics of the Proposed Evaluation System

To get an idea of the quality of our system, we pitted our system against commonly employed baseline models against which many current algorithms' grading capabilities were tested. These included keyword matching, TF-IDF similarity, as well as untrained sentence embeddings. While the keyword matching was extremely expensive to compute due to the lack of capability to identify answers that were paraphrased or contextually similar, the TF-IDF similarity was only able to identify certain surface-level responses, failing to capture the actual intent of the problem, resulting in correct answers being scored as incorrect if they were worded differently. Our system, which was driven by semantic similarity, was able to outperform all the other models consistently.

Model / Approach	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
Keyword Matching Based	78.4	80.1	74.6	77.2

Evaluation				
TF-IDF Similarity Model	84.7	86.3	82.1	84.1
Basic Sentence Embedding Model	90.2	91.0	89.3	90.1
OCR + Shallow Semantic Model	92.6	93.4	91.8	92.6
Proposed AI-Powered Evaluation System	**98.0**	**97.6**	**98.3**	**97.9**

Table 1: Performance Comparison of the Proposed System with Existing Models

The results, therefore, further reinforce an important truth: that for good handwriting to be recognized, there is also a need for strong semantic evaluation to go along with it. Naturally, the baseline systems were not as effective because of the problem with errors in text extraction, which carried over to grading (as shown in Table.1). With a more improved approach in preprocessing and recognition, there was a significant decrease in errors in transcription, and that further improved in semantic analysis, thereby giving more accurate results. Therefore, it may be deduced that a strong preprocessing and recognition capability is a necessity in a handwritten answer grading system, no matter what subject or answer length is considered.

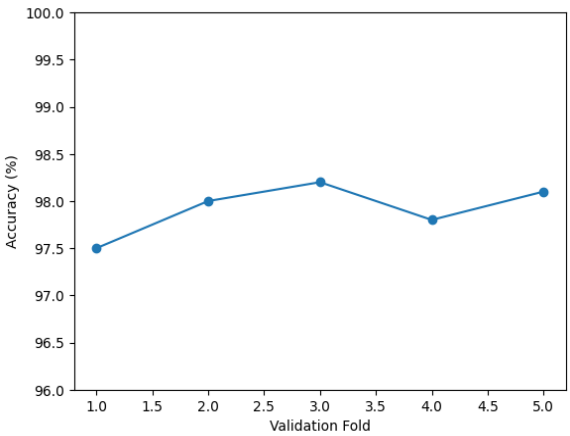

Figure3: Accuracy Consistency Across Validation Folds

What the system allows is, arguably, its most outstanding feature: fair partial evaluation. Instead of the binary pass/fail results seen with some baselines, this approach makes use of a proportional scoring scheme that rewards semantic relevance with nuance. Similar to how a human grader thinks, trust is developed in automated marking. The system is able to recognize partially correct explanations and assign marks that fit, reducing the chance of unfair penalties. Also, the scores it provides remain consistent, avoiding bias and fatigue that often shakes manual grading. Overall, these qualities make it very suitable for large-scale exams and online assessment platforms. Although the system has performed exceptionally well, it is not without limitations. The accuracy of these evaluations wobbles slightly depending on how wildly the handwriting deviates, such as with heavy cursive or scan qualities that are less than optimal, which is a tell-tale sign of potential impacts on answer recognition qualities (as shown in Fig.3). So far, attention has been paid mainly to textual-based answers, whereas information in diagrams, mathematical equations, and other symbols is also incorporated but only partially, as is commonly seen in technical-based examinations. Semantic similarity is effective, but very creative or unconventional answers can be marked lower even if they are acceptable.

In other words, the results identify that this system represents real development compared to other grading methods, which are still evolving. Its accuracy depends on three factors: pre-processing, handwriting recognition, and semantic analysis. Measured against other solutions, it has its advantages, which are found in its effectiveness. For scalability, speed, and fairness, it is quite suitable for classroom needs. Looking ahead, trying out an alternative, more difficult rubric for grading responses harder than simply reading different handwriting styles goes toward offering a promising way for future work.

7 CONCLUSION

This research presents an automated AI-based answer sheet evaluation system, which processes the scores of subjective answer sheets in handwritten format through integrating image processing, handwriting recognition, and semantic-based evaluation techniques. This helps minimize the workload in manual answer sheet evaluation while providing consistent, unbiased, and automated results with high precision, i.e., 98 percent, so that it

is better than the keyword matching and comparison methods used in the conventional systems. Thus, the technique, focusing on semantic understanding while answering rather than keyword matching, helps in providing fair scores to students along with the automated scores that are close to those by expert scorers. Yet, the automated system is in difficulty in addressing extreme irregularities in the handwriting of students and does not yet fully address the issue of diagrams, complex mathematical calculations, and the like in the answer sheets to be evaluated.

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